



Constellation®

LaSalle County Station

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Marseilles, IL 61341
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RA24-024

10 CFR 50.73

July 31, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2024-003-01, Supplement to Technical Specification
(TS) Limiting Condition for Operation (LCO) 3.6.1.3 Completion Time Exceeded

In accordance with 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(vii), Constellation Energy Generation, LLC (CEG) is submitting Licensee Event Report (LER) Number 2024-003-01 for LaSalle County Station, Unit 1.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Ms. Laura Ekern, Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

A handwritten signature in blue ink, appearing to read "Chris Smith".

Christopher J. Smith
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report 2024-003-01

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nureqs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 1



050



052

2. Docket Number

00373

3. Page

1 OF 3

4. Title

Technical Specification (TS) Limiting Condition for Operation (LCO) 3.6.1.3 Completion Time Exceeded

5. Event Date

Month	Day	Year
04	09	2024

6. LER Number

Year	Sequential Number	Revision No.
2024	- 003 -	01

7. Report Date

Month	Day	Year
07	31	2024

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
NA		NA
Facility Name	☐ 052	Docket Number
NA		NA

9. Operating Mode

1

10. Power Level

100 percent

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☒ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER

Licensee Contact

Joseph Reda, Operations Director

Phone Number (Include area code)

(815) 415-2200

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	IG	ISV	G080	Y	NA	NA	NA	NA	NA

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On April 9, 2024, while preparing for the Unit 1 traversing in-core probe (TIP) system shear valve actuating cartridge operability test to meet Technical Specification (TS) Surveillance Requirement (SR) 3.6.1.3.9, it was identified that three of the five installed shear valve squib cartridges had expired in October 2023. The Bases for SR 3.6.1.3.9 requires that the controls that limit shelf life and operating life of the explosive charge be followed, as applicable. Since the five-year expiration date was exceeded for the three identified TIP shear valves, the Bases requirements of SR 3.6.1.3.9 were not met and rendered the three TIP shear valves INOPERABLE. Due to SR 3.6.1.3.9 Bases not being met, TS Limiting Condition for Operation (LCO) 3.6.1.3 is required to be entered. The station exceeded the completion time for TS LCO 3.6.1.3; therefore, this this event is reportable as a condition prohibited by TS per 10 CFR 50.73(a)(2)(i)(B). Since three TIP shear valves were inoperable simultaneously due to the same cause, this event is also reportable per 10 CFR 50.73(a)(2)(vii) as a common cause inoperability of independent trains or channels.

There was no impact to the health and safety of the public or plant personnel because all five squib cartridges were removed and tested successfully. The successful completion of the test demonstrated that the TIP shear valves with the expired squib cartridges were still capable of performing their function with the expired squib cartridges installed.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME LaSalle County Station, Unit 1	☒ 050	2. DOCKET NUMBER 00373	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 01

NARRATIVE**Plant and System Identification**

LaSalle Country Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected system was the traversing in-core probe (TIP) system. The affected components are the 1C51-J004A/D/E TIP system shear valves.

Condition Prior to Event

Unit(s): 1	Date: April 9, 2024	Time: 18:30 CDT
Reactor Mode(s): 1	Mode(s) Name: Power Operation	Power Level: 100 percent

Description

On April 9, 2024, while preparing for the Unit 1 traversing in-core probe (TIP) system shear valve actuating cartridge operability test, it was identified that squib cartridges installed in three out of the five Unit 1 TIP shear valves had expired ('A', 'D', and 'E' trains). The squib cartridges have an expiration date of five years from the time the charge is packed into the firing mechanism. The five-year expiration date is based on the manufacturer's operating experience of squib charges not reliably firing after five years. The previous performance of this test on October 18, 2021, replaced and tested the squib cartridge from the 'B' train TIP shear valve while noting the other four squib cartridges were set to expire in October 2023, prior to the next performance of the test. A separate surveillance performed on January 31, 2023, replaced, and tested the squib cartridge from the 'C' train TIP shear valve.

Upon identification, all applicable Technical Specification (TS) timeclocks were entered, and actions were taken to isolate the affected penetration flow path. The operability test was performed on April 10, 2024, and all five squib charges were replaced, and the removed charges were actuated successfully. The TIP shear valves were declared operable and all applicable TS timeclocks were exited. The successful completion of the test demonstrated that the TIP shear valves with the expired squib cartridges were still capable of performing their function.

Cause

The cause of the event is that the test surveillance did not factor in the expiration date of the squib cartridges for future surveillance test performances. Instead, an alternative tracking method was used to ensure the squib cartridges did not expire prior to the next performance of the operability test, which was ineffective. Additionally, during the last performance of the operability test, the expiration dates of the squib cartridges were not documented in the Corrective Action Program (CAP) to ensure expiration dates were not exceeded.

Reportability and Safety Analysis

The TIP valve assembly contains both a ball valve and a shear valve. When the TIP system is in standby, the system is retracted, and the ball valves are closed to maintain primary containment integrity. When the TIP system is operating and the detectors are in the core, the ball valve is open. In the event of a primary containment isolation signal, the TIP system is automatically withdrawn from the core and the ball valves are closed to maintain containment integrity. The shear valve is only used if a leak occurs when the TIP is beyond the ball valve and the power to the TIP system fails. The shear valve is actuated manually and can cut through and close off the tubing to preserve primary containment integrity. Therefore, both the ball valve and the shear valve are primary containment isolation valves (PCIVs) and must meet the requirements of TS Limiting Condition for Operation (LCO) 3.6.1.3, each PCIV shall be OPERABLE.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 003	REV NO. - 01

NARRATIVE

The TIP shear valve must meet the requirements of TS Surveillance Requirement (SR) 3.6.1.3.9, to remove and test the explosive squib from each shear isolation valve of the TIP system at a frequency that is in accordance with the Surveillance Frequency Control Program (SFCP) which was met. The Bases for SR 3.6.1.3.9 requires that the controls that limit shelf life and operating life of the explosive charge be followed, as applicable. Since the five-year expiration date of October 2023 was exceeded for the three identified TIP shear valves, the Bases requirements of SR 3.6.1.3.9 were not met and rendered the three TIP shear valves INOPERABLE.

Due to SR 3.6.1.3.9 Bases not being met, LCO 3.6.1.3 was required to be entered. Condition C of LCO 3.6.1.3 was applicable because the flow path has only one PCIV (either the ball valve or the shear valve). Condition C.1 requires that the flow path be isolated by at least one closed and de-activated automatic valve, closed manual valve, or blind flange within 4 hours. Also, Condition C.2 must be met to verify the affected penetration flow path is isolated once every 31 days following isolation. The station had exceeded the completion time of 4 hours for Condition C.1 due to the TIP shear valves being inoperable since October 2023. The completion time of Condition C.2 was also not met due to Condition C.1 not being completed when required. Therefore, this event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's TS.

Since three of the five TIP shear valves were inoperable simultaneously due to the same cause, this event is also reportable per 10 CFR 50.73(a)(2)(vii) as a common cause inoperability of independent trains or channels.

Corrective Actions

Corrective actions taken in response to the conditions were:

- Declared the affected TIP shear valves inoperable and entered required TS timeclocks and actions.
- Replaced the squib charges in all five TIP shear valves.
- Test fired the five removed squib charges and confirmed they all fired successfully.
- Declared the affected TIP shear valves operable and exited applicable TS timeclocks and actions.
- Performed an extent of condition review for Unit 2 that identified the TIP shear valve squib cartridges were in compliance.
- Revise test instructions to add a step requiring the documentation in CAP of the test completion, squib cartridge expiration dates, and recommendations from testing for tracking purposes.
- Revise the test frequency to limit the allowed grace time for the late performance date.

Previous Occurrences

No previous occurrences.

Component Failure Data

Manufacturer: General Electric
Device: Unit 1 TIP System Shear Valves
Component ID: 1C51-J004A/D/E TIP Drive Guide Tube Valve Assembly